

DOCUMENTATION

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Subject:

Information Security

AI-Based Adaptive Login Defense System

Introduction:

In today's digital landscape, traditional "static" login systems, which only check whether a password is correct, are no longer sufficient. This project, **AI-Based Adaptive Login Defense**, adds an intelligent security layer. Instead of a simple binary check, this system monitors behavioral patterns and session risk. By analyzing input speed and failure frequency, the system dynamically scales its defense, eventually implementing a permanent lockout for identified attackers.

Problem Statement:

Standard authentication systems suffer from several critical vulnerabilities:

- **Binary Logic:** They treat a fast-typing bot and a slow-typing human the same way if the password is correct.
- **Brute Force Vulnerability:** They often allow unlimited attempts or simple session-based lockouts that are easily bypassed.
- **Static Defense:** They do not escalate security measures based on the perceived threat level of a specific session.

Objectives:

- **Behavioral Analysis:** To measure "Detection Delays" and input rhythms to distinguish between humans and automated scripts.
- **Risk-Based Escalation:** To use a numerical Risk Score (0–100) that dictates system response.
- **Permanent Threat Neutralization:** To implement a "Hard Lock" state once a source is confirmed as an attacker.

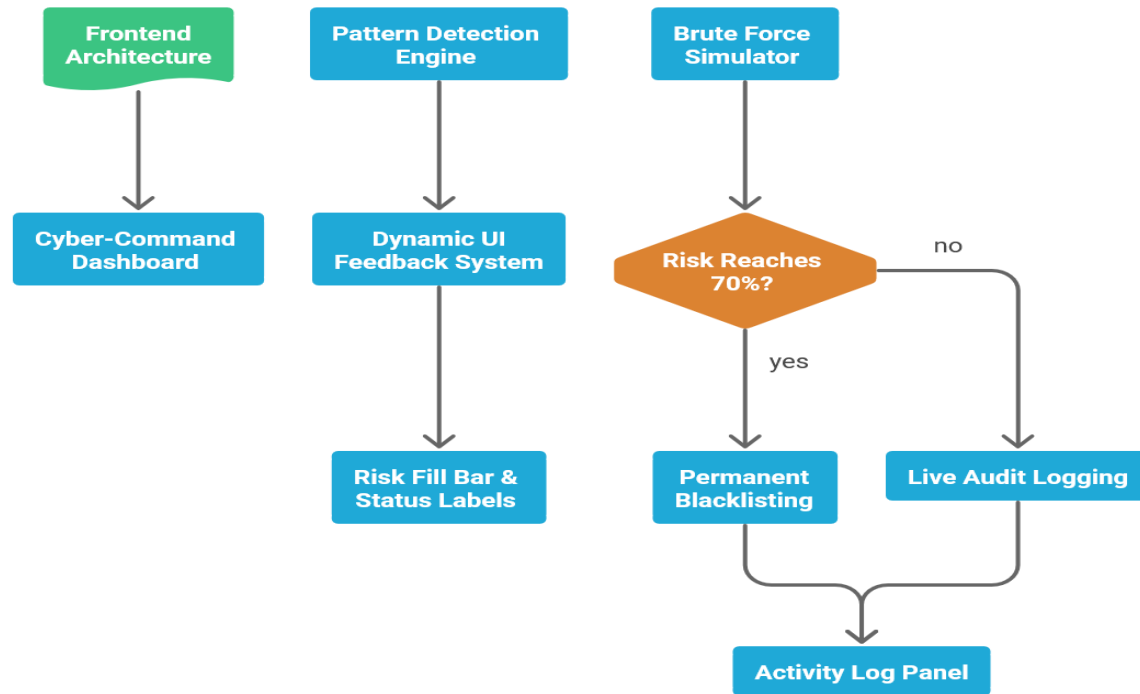
Methodology (Working)

The methodology describes the logical framework and strategy used to build the defense system.

- **Risk-Based Security Model:** Instead of a simple "yes/no" authentication, the system employs a mathematical risk score (0-100) to determine the trust level of a session.
- **Behavioral Pattern Analysis:** The system identifies threats by analyzing non-human behaviors, such as the rapid execution of login attempts (simulated at 0.3s intervals) which indicates automation.
- **Escalating Defense Strategy:** The system logic is designed to become more restrictive as the risk score increases, moving from "Safe" to "Suspicious" and finally "Threat Detected."
- **Zero-Trust Session Management:** Every session starts with a baseline risk, and trust must be maintained through "human-like" interaction.
- **State-Persistent Lockout:** The logic uses a global state variable to ensure that once a high-risk threshold is crossed, the system enters a "Hard Lock" state that persists even if the user stops the attack.

Implementation Detail:

Implementation Details of Security System



Made with  Napkin

Results & Observation:

During testing, the following behaviors were observed:

- **Human Simulation:** Legitimate attempts resulted in low risk scores (under 30), maintaining a "Safe" status.
- **Brute Force Simulation:** When the automated attack simulation was triggered, the system identified the malicious pattern within 4–5 attempts.
- **System Lockdown:** Upon reaching the 75% risk threshold, the system successfully triggered a "Permanent Lock," disabling the login button and blacklisting the session.

Scenario	Behavior	Risk Score	System Action
Legitimate User	Correct password, slow input	0 - 20	Access Granted
Suspicious User	Multiple failed attempts	30 - 70	Status: Suspicious
Automated Bot	Fast, repetitive failures	75+	PERMANENT LOCKDOWN

Conclusion:

The **AI-Based Adaptive Login Defense** project demonstrates that security must be proactive and context-aware. By integrating a risk-based engine that tracks behavior rather than just strings of text, we have created a resilient shield against modern brute-force attacks. The implementation of a permanent lockout ensures that once a threat is identified, it is neutralized for the remainder of the session.